

A 5th-Order Nilpotent Algebraic Framework for Regularized Division by Zero: Axiomatic Construction and Physical Applications

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July 13, 2026

Abstract

We introduce a 5th-order nilpotent algebraic extension of the real numbers, denoted as j -Algebra, designed to consistently formalize division by zero for non-zero numerators while preventing classical algebraic collapse. By adjoining a nilpotent symbolic element j satisfying $j^5 = 0$ (with $j^k \neq 0$ for $1 \leq k < 5$), we establish the foundational relation $x/0 = xj$ for all $x \in \mathbb{R} \setminus \{0\}$, distinct from $0/0 = \infty$. We construct the full 5-component element structure, arithmetic operations, exact multiplicative inverses via truncated Neumann series, a 5×5 matrix representation, and 4th-order Taylor series extensions. Finally, we apply this framework to classical electrodynamics and field theory, demonstrating that point-source field singularities, self-force divergences, and stress-energy tensors $T_{\mu\nu}$ are regularized into unobservable nilpotent ghost sectors without violating conservation laws or relativistic covariance.

1 Introduction

Division by zero in classical field theories and real analysis typically leads to undefined operations or ill-defined physical infinities. Traditional approaches either ban division by zero entirely or handle it through distribution theory or renormalization protocols.

In this paper, we construct a closed algebraic system that formalizes division by zero as a 5th-order nilpotent extension of the real line. By adjoining a generator j obeying $j^5 = 0$, singular operations $x/0$ ($x \neq 0$) are mapped to scaled nilpotent terms. This structure isolates divergent behavior into an unobservable "ghost sector" up to order 4 while preserving standard arithmetic for real quantities.

2 Axiomatic Foundations

2.1 Core Definitions

Let $\mathbb{R}$ denote the set of real numbers. We extend $\mathbb{R}$ to a commutative ring $\mathbb{R}[j]/\langle j^5 \rangle$ by adjoining an element j subject to the following postulates:

Definition 2.1 (Division by Zero). For any non-zero real number $x \in \mathbb{R} \setminus \{0\}$:

$$\frac{x}{0} = xj \tag{1}$$

Postulate 2.1 (Indeterminate Form). The ratio $0/0$ is explicitly distinguished from $x/0$ and defined as absolute infinity:

$$\frac{0}{0} = \infty \tag{2}$$

Postulate 2.2 (5th-Order Nilpotency). The generator j obeys a 5th-order nilpotency condition:

$$j^5 = 0 \quad \text{with} \quad j^k \neq 0 \quad \text{for} \quad 1 \leq k \leq 4 \quad (3)$$

Postulate 2.3 (Annihilation by Zero). Multiplication of j by the real zero satisfies $0 \cdot j = 0$.

Postulate 2.4 (Observability Principle). For any element $z = a_0 + a_1j + a_2j^2 + a_3j^3 + a_4j^4 \in \mathbb{R}[j]/\langle j^5 \rangle$, its physically observable value is given strictly by its real projection:

$$\mathcal{O}(z) = a_0 \in \mathbb{R} \quad (4)$$

Nilpotent components a_kj^k ($k \geq 1$) represent internal mathematical degrees of freedom and are unobservable in physical measurements.

2.2 General Elements and Arithmetic

A general element $z \in \mathbb{R}[j]/\langle j^5 \rangle$ is a linear combination over the basis $\{1, j, j^2, j^3, j^4\}$:

$$z = \sum_{k=0}^4 a_k j^k = a_0 + a_1j + a_2j^2 + a_3j^3 + a_4j^4 \quad (5)$$

For $z = \sum_{k=0}^4 a_k j^k$ and $w = \sum_{k=0}^4 b_k j^k$:

1. **Addition:** $z + w = \sum_{k=0}^4 (a_k + b_k) j^k$
2. **Multiplication:** $z \cdot w = \sum_{k=0}^4 c_k j^k$ where the coefficients are given by the discrete convolution:

$$c_k = \sum_{m=0}^k a_m b_{k-m} \quad (6)$$

All products with powers j^n ($n \geq 5$) vanish identically.

2.3 Multiplicative Inverses

For any element $z = a_0 + N$ with $a_0 \neq 0$ and $N = \sum_{k=1}^4 a_k j^k$, N is strictly nilpotent ($N^5 = 0$). The exact multiplicative inverse z^{-1} is derived via the truncated Neumann series:

$$z^{-1} = \frac{1}{a_0} \sum_{m=0}^4 \left(-\frac{N}{a_0} \right)^m = \frac{1}{a_0} \left[1 - \frac{N}{a_0} + \left(\frac{N}{a_0} \right)^2 - \left(\frac{N}{a_0} \right)^3 + \left(\frac{N}{a_0} \right)^4 \right] \quad (7)$$

2.4 Matrix Representation

The algebra $\mathbb{R}[j]/\langle j^5 \rangle$ is isomorphic to a subalgebra of 5×5 real matrices, where j maps to a Jordan shift block:

$$j \mapsto \mathbf{J} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad z \mapsto \begin{pmatrix} a_0 & a_1 & a_2 & a_3 & a_4 \\ 0 & a_0 & a_1 & a_2 & a_3 \\ 0 & 0 & a_0 & a_1 & a_2 \\ 0 & 0 & 0 & a_0 & a_1 \\ 0 & 0 & 0 & 0 & a_0 \end{pmatrix} \quad (8)$$

Matrix multiplication directly verifies $j^5 = \mathbf{0}_{5 \times 5}$.

3 Calculus and Analytic Extensions

3.1 Functions of j -Numbers

For any smooth, real-analytic function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(a_0 + N)$ expands exactly up to the 4th derivative:

Proposition 3.1 (Analytic Extension).

$$f(a_0 + N) = f(a_0) + f'(a_0)N + \frac{f''(a_0)}{2!}N^2 + \frac{f'''(a_0)}{3!}N^3 + \frac{f^{(4)}(a_0)}{4!}N^4 \quad (9)$$

4 Physical Applications

4.1 Regularization of Point Source Singularities

Consider the electrostatic field of a point charge q at $\mathbf{r} = 0$:

$$\mathbf{E}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0 r^2} \hat{\mathbf{r}} \implies \mathbf{E}(0) = \frac{q}{4\pi\epsilon_0} j \hat{\mathbf{r}} \quad (10)$$

Under $j^5 = 0$, the energy density $u(0) = \frac{1}{2}\epsilon_0 |\mathbf{E}(0)|^2 = \frac{q^2}{32\pi^2\epsilon_0} j^2 \hat{\mathbf{r}} \cdot \hat{\mathbf{r}}$ contains a j^2 component. While non-zero algebraically, applying the Observability Principle yields:

$$\mathcal{O}(u(0)) = \mathcal{O}\left(\frac{q^2}{32\pi^2\epsilon_0} j^2\right) = 0 \quad (11)$$

Thus, infinite self-energy divergences vanish in all physical measurements.

4.2 Covariant Stress-Energy Tensor

Expanding the electromagnetic field tensor $F_{\mu\nu} = F_{\mu\nu}^{(\text{real})} + jF_{\mu\nu}^{(\text{sing})}$, the stress-energy tensor $T_{\mu\nu} \propto F_{\mu\alpha}F_{\nu}{}^{\alpha}$ yields components up to j^2 . Under physical projection $\mathcal{O}(T_{\mu\nu})$, all higher-order nilpotent terms j^k ($k \geq 1$) drop out, keeping the gravitational coupling $G_{\mu\nu} = \frac{8\pi G}{c^4} \mathcal{O}(T_{\mu\nu})$ strictly finite.

5 Conclusion

The 5th-order nilpotent j -algebra provides an airtight mathematical foundation for regularizing division by zero. By incorporating terms up to j^4 and employing the Observability Principle, singularities in classical electrodynamics and general relativity are cleanly isolated into ghost sectors without generating non-real physical observables.

References

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